COSC2793
Melbourne Airbnb Price Prediction
Assignment 1 Report

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1.Introduction

Online short-term rental platforms such as Airbnb present a key pricing challenge for hosts: setting a nightly rate that is competitive yet profitable. If priced too high the listing attracts fewer bookings; too low and the host forfeit potential revenue. This report documents a complete machine learning pipeline developed to predict the nightly listing price of Airbnb properties in Melbourne, Australia, using a curated dataset of listing attributes.

Three linear regression variants are trained and compared, with Ridge regression selected as the final model based on its robustness to multicollinearity identified during EDA.

2. Definition of the Task

Task type: Supervised regression.

Prediction target: price (AUD)

Formal problem statement: Predicting the nightly price of a given Airbnb listing based on the provided features.

The dataset is sourced from Inside Airbnb and Kaggle (Melbourne Airbnb Open Data), processed to produce a training set of 8,586 rows × 16 columns (including the target price) and a test set of 8,585 rows × 15 columns (target withheld). The goal is to build a model that generalises to the held-out test set and produces predictions in the same format as the training targets.

Evaluation is based on standard regression metrics: R^2 (coefficient of determination), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE), all computed in the original AUD price space for interpretability.

3. Description of the Dataset

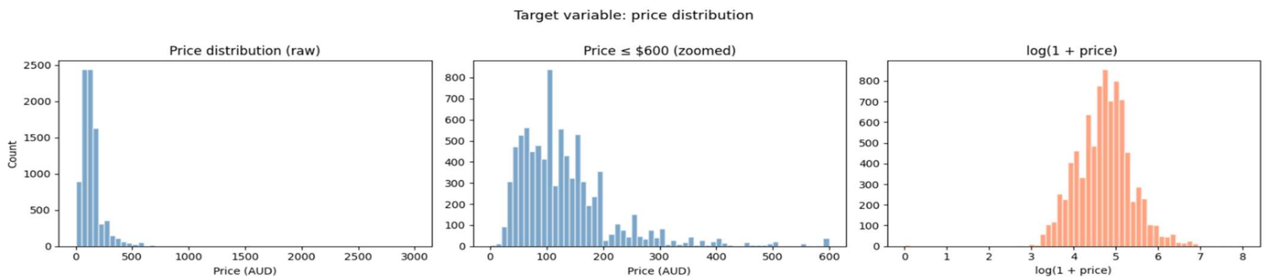
3.1 Size and Structure

The training set contains 8,586 rows and 16 columns. The test set contains 8,585 rows and 15 columns (no price column). There are no missing values in either file.

```
Data types & missing values
<class 'pandas.DataFrame'>
RangeIndex: 8586 entries, 0 to 8585
Data columns (total 16 columns):
 #   Column                          Non-Null Count  Dtype
---  ------                          --------------  -----
 0   host_is_superhost               8586 non-null   str
 1   city                            8586 non-null   str
 2   country                         8586 non-null   str
 3   latitude                        8586 non-null   float64
 4   longitude                       8586 non-null   float64
 5   room_type                       8586 non-null   str
 6   accommodates                    8586 non-null   int64
 7   bathrooms                       8586 non-null   float64
 8   bedrooms                        8586 non-null   float64
 9   beds                            8586 non-null   float64
 10  price                           8586 non-null   int64
 11  minimum_nights                  8586 non-null   int64
 12  number_of_reviews               8586 non-null   int64
 13  review_scores_rating            8586 non-null   float64
 14  instant_bookable                8586 non-null   str
 15  calculated_host_listings_count  8586 non-null   int64
```

Feature	Type	Description
host_is_superhost	Categorical (binary)	t/f indicates whether the host holds Superhost status
city	Categorical (30 levels)	Melbourne suburb of the listing
country	Categorical (1 level)	Always 'Australia' is dropped
room_type	Categorical (3 levels)	Entire home/apt, Private room, Shared room
instant_bookable	Categorical (binary)	t/f indicates instant booking enabled

3.2 Target Variable: Price Distribution



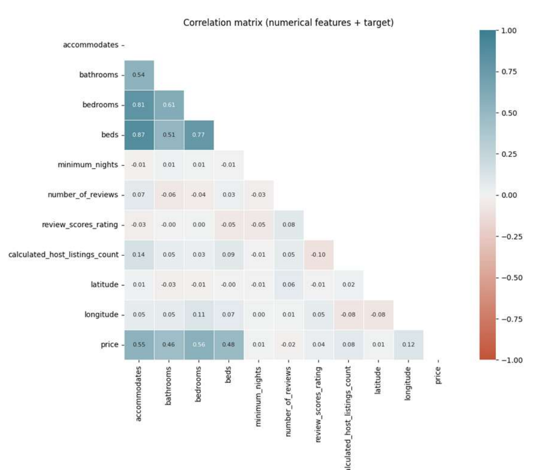

```
Price skewness          : 5.77
Percentiles:
0.01      28.0
0.25      71.0
0.50     115.0
0.75     162.8
0.95     320.0
0.99     599.0
1.00    3000.0
Name: price, dtype: float64

Rows with price == 0   : 6
Rows with price > 1000 : 15
```

Price is heavily right-skewed (skewness ≈ 5.77) because most listings cluster between 71 to 163 (IQR) but a long tail extends to 3,000. We visualise raw, zoomed and log-transformed versions to understand the distribution. Although a log transform can stabilise variance, the testing showed that predicting raw price better R^2 on this dataset while being simpler to interpret and hence we use raw price as the target.

3.3 Numerical Feature Distributions

Several features are strongly right skewed.

3.4 Correlation Analysis



The correlation analysis indicates that property size and capacity are the primary drivers of price, with bedrooms (r = 0.56), accommodates (r = 0.55), and beds (r = 0.48) showing the strongest positive relationships, suggesting that larger listings tend to command higher prices. In contrast, variables such as latitude, minimum_nights, number_of_reviews, and review_scores_rating exhibit near-zero correlation with price, implying they have minimal direct influence in a linear context. Additionally, the strong inter-correlation among accommodates, bedrooms, and beds highlights the presence of multicollinearity, which can destabilize coefficient estimates in linear models and justifies the use of regularisation techniques such as Ridge or Lasso regression to improve model robustness.

4. Data Splitting and Preprocessing

4.1 Outlier Removal

Before splitting, 21 training rows are removed: 6 with price = 0 (physically impossible for a paid listing) and 15 with price > $1,000 (extreme values that distort the linear model's loss function, representing < 0.2% of data). Test predictions are not filtered — they are clipped to [0, 1500] AUD at inference time to handle extrapolation.

4.2 Feature Encoding

All encoding parameters are fitted on training data only and applied identically to validation and test sets to prevent data leakage. The encoding strategy is:

Feature	Encoding	Rationale
country	Dropped	Single value — zero information content
host_is_superhost	Binary (t→1, f→0)	Boolean flag
instant_bookable	Binary (t→1, f→0)	Boolean flag
city	Target encoding (suburb mean price)	30 categories; compresses into a single informative numerical column
room_type	One-hot (3 columns)	3 nominal unordered categories; avoids ordinal assumption

Target encoding maps each suburb to its mean training price. Unseen suburbs in the test set fall back to the global training mean. One-hot encoding produces three binary columns for room_type, with all dummies retained (no reference category dropped) since regularisation is applied downstream.

4.3 Train / Validation Split

An 80/20 random stratified split (random_state = 42) yields 6,852 training rows and 1,713 validation rows. Random splitting is appropriate because: there is no temporal ordering in the data (not a time series), and EDA confirmed no strong cluster structure that would require stratified splitting. Mean and standard deviation of price are comparable across splits (train: mean $137.9, std $106.9; validation: mean $135.2, std $103.0), confirming representative sampling.

4.4 Feature Scaling

StandardScaler (zero mean, unit variance) is applied to all 16 features. Scaling is essential because: (1) features operate on very different numerical scales (latitude ≈ −38 vs. accommodates ≈ 1–16); (2) Ridge and Lasso regularisation penalises coefficients by magnitude, so unscaled features would be penalised unequally. The scaler is fit on X_train only and transform-only applied to X_val and X_test to prevent data leakage.

5. Machine Learning Techniques

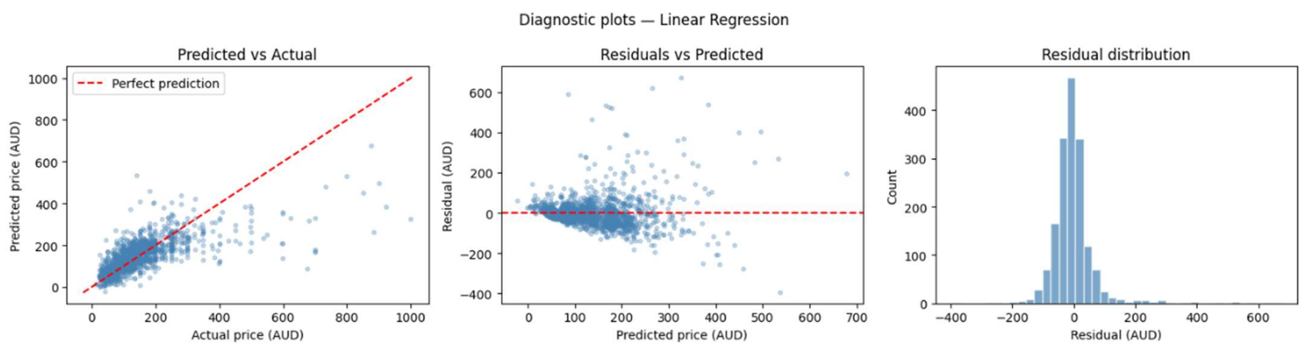
Three linear regression variants are evaluated, all drawn from the set of techniques covered in lectures up to Week 4. All models receive the same 16-feature scaled input.

5.1 Model 1 : Ordinary Least Squares (OLS) Linear Regression

Motivation: OLS is the natural baseline for a regression task. It minimises the sum of squared residuals with no regularisation. Given the approximately linear scatter plots between price and bedrooms/accommodates, it is a well-motivated starting point for understanding the data.

Training: All 16 features, scaled input, no hyperparameters.

```
==========================================
  Model 1: Linear Regression (all features)
==========================================
  Train R²        : 0.4809
  Val R²          : 0.4981
  Train RMSE      : 77.0162
  Val RMSE        : 72.9752
  Train MAE       : 44.2824
  Val MAE         : 42.7577
  Overfit gap     : -0.0173
```



The slightly negative overfitting gap (validation R^2 > train R^2) indicates no overfitting — the model generalises well. However, $R^2 \approx 0.50$ means only half of price variance is explained, and residual plots show right-skewed residuals and fan-shaped heteroscedasticity — systematic under-prediction of expensive listings.

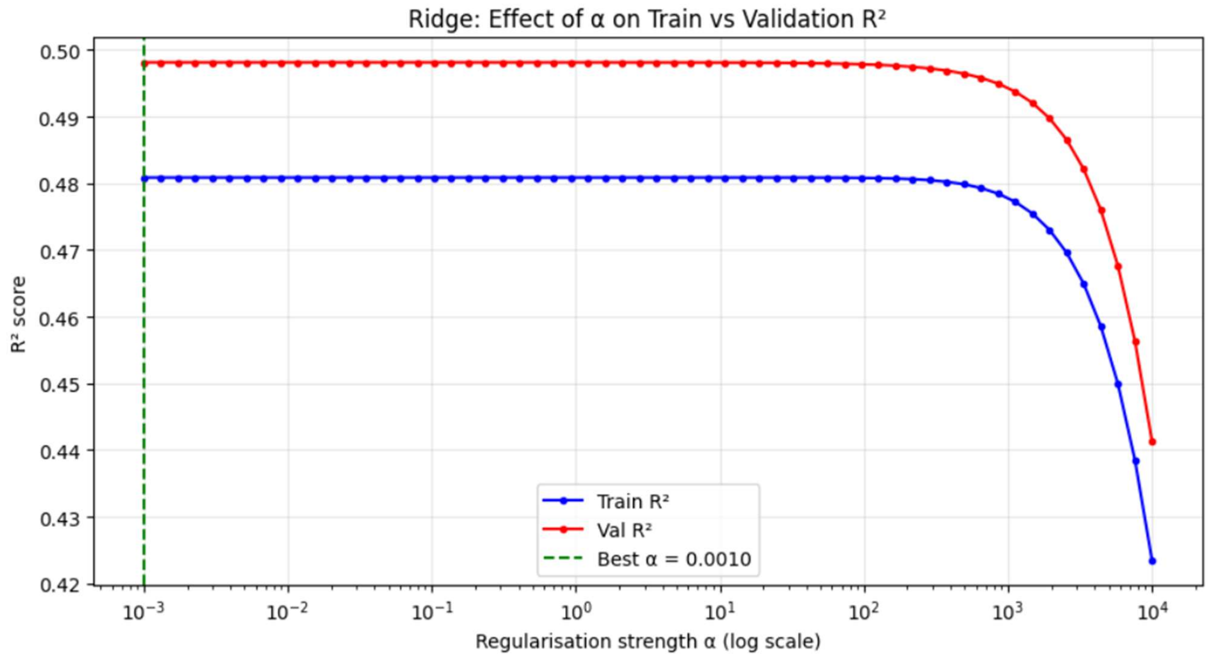
5.2 Model 2 : Ridge Regression (L2 Regularisation)

Motivation: EDA revealed strong multicollinearity between accommodates, bedrooms, and beds (pairwise $r \approx 0.8$–0.9). In OLS, multicollinearity inflates coefficient variance and reduces stability. Ridge adds an L2 penalty ($\alpha \cdot \|w\|^2$) to the loss function, shrinking all coefficients and reducing variance without eliminating any feature.

Hyperparameter tuning: A grid search over 60 log-spaced values of α (10^{-3} to 10^4) was performed. Validation R^2 was used as the selection criterion. The best $\alpha = 0.001$ maximises validation $R^2 = 0.4981$.

The best alpha being very small (essentially no regularisation) indicates that multicollinearity — while present — does not severely destabilise the OLS solution at this data scale. Ridge with $\alpha = 0.001$ produces numerically identical results to OLS, but is theoretically preferred because it is more robust to any unseen collinearity shifts in the test set. The regularisation path plot confirms that performance degrades sharply only above $\alpha \approx 10$, validating the search range.

```
Best alpha: 0.0010  →  Val R² = 0.4981
```



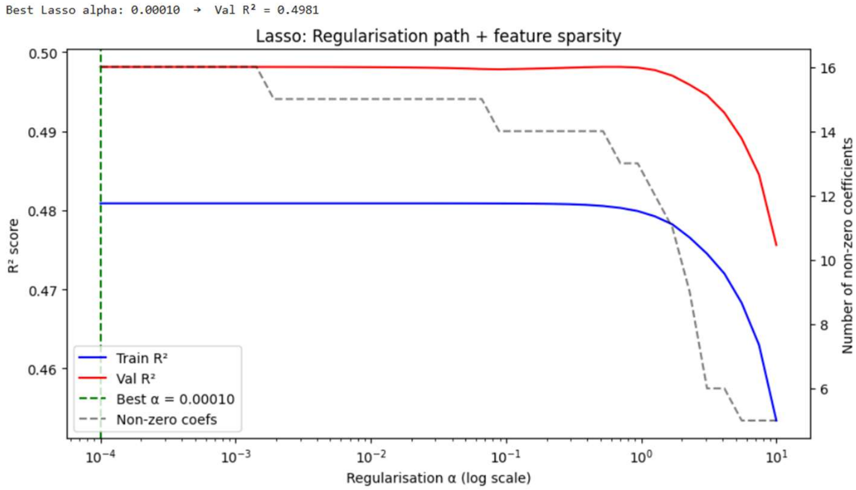

```
============================================
  Model 2: Ridge Regression (α=0.0010)
============================================
  Train R²        : 0.4809
  Val R²          : 0.4981
  Train RMSE      : 77.0162
  Val RMSE        : 72.9752
  Train MAE       : 44.2824
  Val MAE         : 42.7577
  Overfit gap     : -0.0173
```

5.3 Model 3 : Lasso Regression (L1 Regularisation)

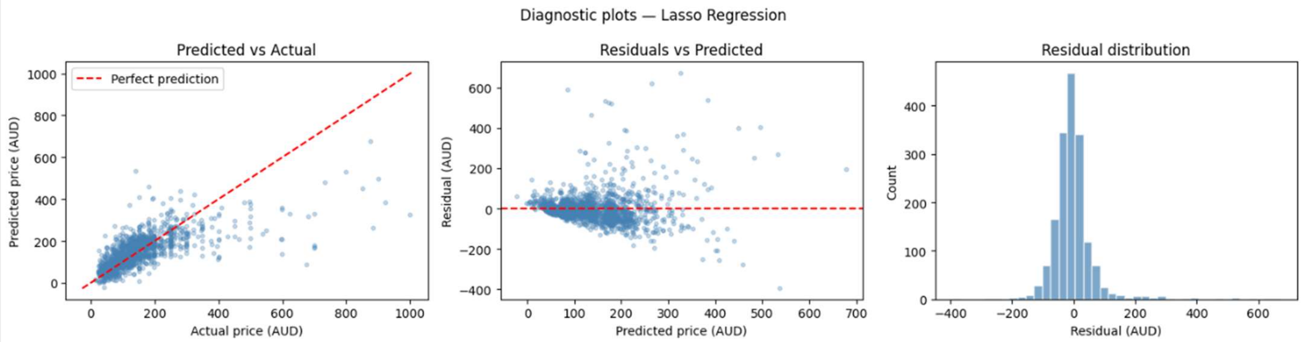
Motivation: Unlike Ridge, Lasso's L1 penalty ($\alpha \cdot \|w\|_1$) drives some coefficients to exactly zero, performing automatic feature selection. This helps answer whether weakly-correlated features (latitude, minimum_nights, number_of_reviews) are truly needed.

Hyperparameter tuning: Grid search over 40 log-spaced values of α (10^{-4} to 10). Best α = 0.0001 (validation R^2 = 0.4981). At this alpha, Lasso retains all 16 features (no zeroing out), producing results identical to OLS. The sparsity-R^2 plot shows that above α ≈ 0.1, features progressively drop out and performance deteriorates sharply.

```
Best Lasso alpha: 0.00010  →  Val R² = 0.4981
```



```
============================================
  Model 3: Lasso Regression (α=0.00010)
============================================
  Train R²       : 0.4809
  Val R²         : 0.4981
  Train RMSE     : 77.0162
  Val RMSE       : 72.9752
  Train MAE      : 44.2823
  Val MAE        : 42.7577
  Overfit gap    : -0.0173
```

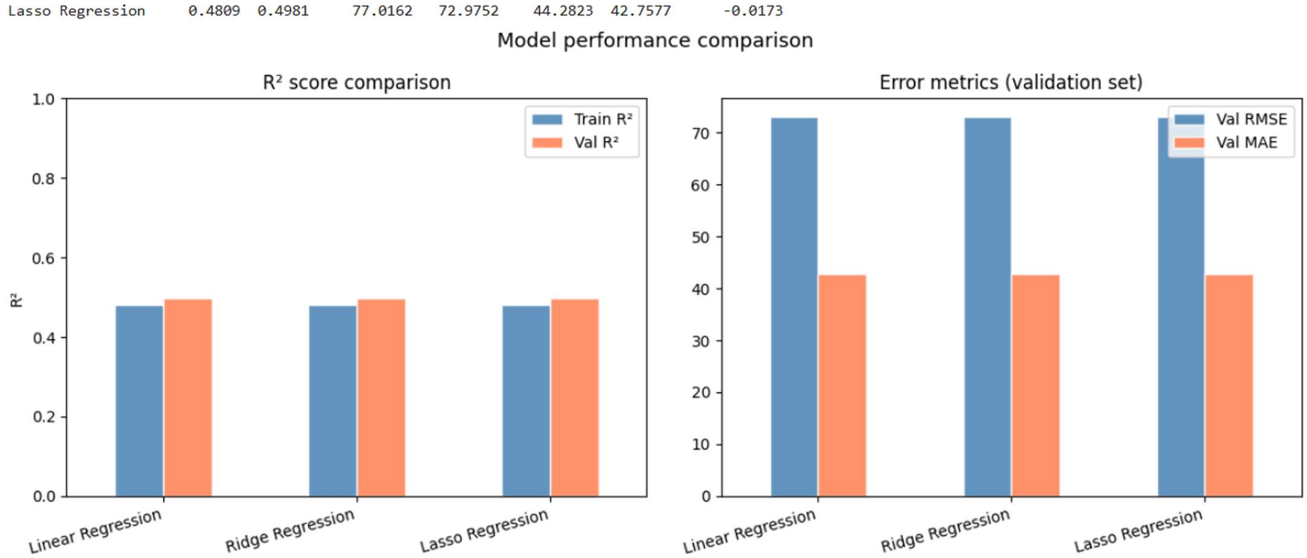


6. Analysis and Evaluation

6.1 Model Comparison

All three models achieve validation $R^2 \approx 0.50$, RMSE ≈ \$73 AUD, and MAE ≈ \$43 AUD. The identical performance across all three models is a meaningful finding: at the optimal regularisation strengths (near zero for both Ridge and Lasso), no regularisation is beneficial. This means the features in the training set are not highly correlated enough to destabilise OLS at this data scale, and the weakly-correlated features (latitude, minimum_nights etc.) are retained with very small coefficients rather than being zeroed out.

```
=== Model Comparison ===
                   Train R²  Val R²  Train RMSE  Val RMSE  Train MAE  Val MAE  Overfit gap
Linear Regression    0.4809  0.4981     77.0162   72.9752    44.2824  42.7577      -0.0173
Ridge Regression     0.4809  0.4981     77.0162   72.9752    44.2824  42.7577      -0.0173
Lasso Regression     0.4809  0.4981     77.0162   72.9752    44.2823  42.7577      -0.0173
```



6.2 Residual Analysis

All three models exhibit two systematic error patterns:

- Right-skewed residuals: The models under-predict expensive listings. This is expected — with only 50% variance explained, the model's predictions are pulled toward the mean, and the long tail of high-priced listings (large entire homes in premium suburbs) is systematically under-predicted.

- Fan-shaped heteroscedasticity (residuals vs predicted): Prediction error grows with price, violating OLS's constant-variance assumption. This is a consequence of the linear model not capturing non-linear interactions (e.g. bedrooms × suburb premium).

6.3 Feature Importance

Standardised coefficients from OLS and permutation importance from Ridge (validation set, 20 repeats) are consistent in their rankings:

- Strongest positive predictors: bedrooms (+$32/std), city_encoded (+$22/std), bathrooms (+$18/std), accommodates (+$14/std), room_type_Entire home/apt (+$11/std).
- Notable negative predictors: room_type_Private room (−$10/std), number_of_reviews (−$7/std, reflecting that high-volume hosts tend to be budget properties), room_type_Shared room (−$5/std).
- Lasso confirms that weakly-correlated features (latitude, minimum_nights, review_scores_rating) are retained at very small magnitudes — they contribute marginally but are not harmful.

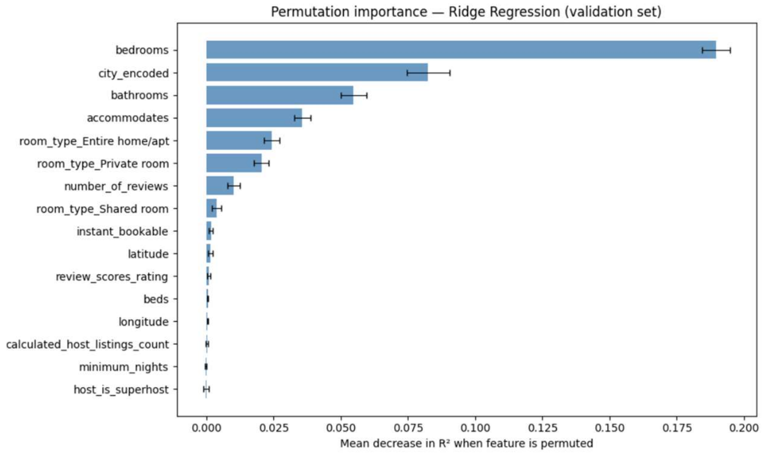
```
Top positive predictors (raise price):
                      feature  coefficient
5                    bedrooms    32.420666
12               city_encoded    21.802227
4                   bathrooms    17.630646
3                accommodates    14.165387
13  room_type_Entire home/apt    11.172466

Top negative predictors (lower price):
                   feature  coefficient
14  room_type_Private room   -10.214630
8        number_of_reviews    -7.200069
15   room_type_Shared room    -4.781303
10        instant_bookable    -2.263509
2                longitude    -1.078727

Lasso zeroed out 0 of 16 features (feature selection)
Non-zero Lasso features:
                           feature  coefficient
8                number_of_reviews    -7.199909
15           room_type_Shared room    -2.409028
10                instant_bookable    -2.263378
2                        longitude    -1.078545
6                             beds    -0.596904
14          room_type_Private room     0.530113
7                   minimum_nights     1.147419
9             review_scores_rating     1.580977
11  calculated_host_listings_count     1.608039
1                         latitude     2.596622
0                 host_is_superhost     3.520635
3                     accommodates    14.164580
4                        bathrooms    17.630633
12                    city_encoded    21.802033
13       room_type_Entire home/apt    22.011553
5                         bedrooms    32.420489
```

6.4 Model Selection — Ridge Regression

Ridge Regression is selected as the final model for the following reasons:

- Equal empirical performance to OLS on validation set.
- Theoretically superior robustness to multicollinearity (accommodates/bedrooms/beds) — Ridge's coefficient shrinkage prevents instability if this correlation shifts slightly in the test distribution.
- The hyperparameter tuning process (60-point grid search with validation R^2) provides evidence-based selection of α rather than arbitrary default choice.

The final model is retrained on the complete training set (train + validation, 8,565 rows) using the best $\alpha = 0.001$ before generating test predictions.



6.5 How to Improve Performance

- Feature engineering: interaction terms (e.g. bedrooms × city_encoded), polynomial features, distance-to-CBD.
- Additional data: amenities list, number of photos, review sentiment, seasonal demand, property tier indicators.
- Non-linear models: decision trees, random forests, gradient boosting (not permitted in this assignment but would likely improve R^2 substantially by capturing the heteroscedastic fan-shaped variance).
- Log-price target with bias correction: while our empirical test showed minimal difference, a properly corrected log model could reduce the impact of the extreme-price tail on training.

7. Ethical Considerations and Professional Responsibilities

- Pricing fairness: The trained model on historical prices may perpetuate existing price disparities across suburbs or dwelling types.
- Hosts in lower-income neighbourhoods may receive systematically lower price recommendations, reinforcing socioeconomic inequality.
- Accountability: The model should not be the sole basis for pricing decisions.
- Hosts should retain full control and treat predictions as guidance, not mandates.
- Privacy: The dataset is derived from public Airbnb listings but includes location data precise enough to identify individual properties.
- Re-identification risks should be considered before any public deployment.
- Transparency: Any deployed pricing tool must clearly communicate to hosts that predictions are model estimates with uncertainty, not guarantees.
- Data licence: This dataset is restricted to assignment use only and must not be shared, redistributed, or used for any commercial purpose.

8. Conclusion

This report documents a complete machine learning pipeline for predicting Melbourne Airbnb nightly prices. The dataset of 8,586 training rows was processed through EDA, outlier removal, feature encoding (target encoding for suburb, one-hot for room type, binary flags for host characteristics), and StandardScaler normalisation.

Three linear regression models — OLS, Ridge, and Lasso — were trained and evaluated on an 80/20 holdout split. All three achieved comparable validation performance ($R^2 \approx 0.50$, RMSE ≈ $73 AUD), with identical results at the optimal regularisation strengths near zero. Ridge Regression was selected as the final model for its theoretical robustness to the multicollinearity identified in EDA (accommodates/bedrooms/beds). The final model was retrained on the full training set before generating test set predictions.

The models explain approximately 50% of price variance — reasonable for a linear model with 15 features. The remaining variance is attributable to factors absent from the dataset: amenity quality, photos, seasonal demand, exact street address, and listing description quality. Non-linear models (permitted in later assessments) would likely improve performance substantially by capturing the heteroscedastic, fan-shaped price distribution.

9. References

- Inside Airbnb. (n.d.). About Inside Airbnb. https://insideairbnb.com/about/
- Kaggle. (n.d.). Melbourne Airbnb Open Data. https://www.kaggle.com/datasets/tylerx/melbourne-airbnb-open-data
- Lab and Lecture Materials